Worked Answer Key: Practice Problems, AP Precalc Units 1 and 2

Unit 1: Polynomial and rational functions

Rates of change

W1. Average rate of change of $f(x) = 3x - 7$ on $[2, 6]$.

- $f(6) = 18 - 7 = 11$ and $f(2) = 6 - 7 = -1$.
- $\dfrac{f(6) - f(2)}{6 - 2} = \dfrac{11 - (-1)}{4} = \dfrac{12}{4} = \boxed{3}$
- Check: a line's average rate of change is always its slope, and the slope here is 3.

W2. Average rate of change of $f(x) = x^2$ on $[0, 2]$ and $[2, 4]$.

- $[0, 2]$: $\dfrac{f(2) - f(0)}{2 - 0} = \dfrac{4 - 0}{2} = 2$
- $[2, 4]$: $\dfrac{f(4) - f(2)}{4 - 2} = \dfrac{16 - 4}{2} = 6$
- The second is bigger. The rate of change is increasing, so the graph gets steeper as x grows: **concave up**.

1. Table: $f = 1, 4, 9, 16, 25$ at $x = 0, \ldots, 4$.

- (a) Subtract consecutive outputs: $4 - 1 = 3,\ 9 - 4 = 5,\ 16 - 9 = 7,\ 25 - 16 = 9$. The changes are **3, 5, 7, 9**.
- (b) The changes are getting bigger, so the rate of change is **increasing** and f is **concave up**.
- (c) Take differences of the differences: $5 - 3 = 2$, $7 - 5 = 2$, $9 - 7 = 2$. The second differences are constant, so f is **quadratic**. (It's $(x + 1)^2$.)

2. $f(x) = -x^2 + 6x$ on $[0, 2]$, $[2, 4]$, $[4, 6]$.

- Values: $f(0) = 0,\ f(2) = -4 + 12 = 8,\ f(4) = -16 + 24 = 8,\ f(6) = -36 + 36 = 0$.
- $[0, 2]$: $\frac{8-0}{2} = 4$. $[2, 4]$: $\frac{8-8}{2} = 0$. $[4, 6]$: $\frac{0-8}{2} = -4$.
- The rates **4, 0, −4** keep decreasing, so f is **concave down**. It rises, levels off at the top, then falls.

3. $h(t) = -16t^2 + 64t$ on $[1, 3]$.

- $h(1) = -16 + 64 = 48$ and $h(3) = -144 + 192 = 48$.
- $\dfrac{48 - 48}{3 - 1} = \boxed{0 \text{ ft/s}}$
- **Meaning:** the ball is at the same height (48 ft) at $t = 1$ and $t = 3$, so its net change in height over that interval is zero.
- **Not the meaning:** the ball didn't sit still. It went up to its peak (at $t = 2$) and came back down. An average hides what happens in between.

4. Increasing and concave down.

- Increasing means the rate of change is **positive**.
- Concave down means the rate of change is **decreasing**.
- Together: the function is still going up, but more slowly over time.
- Examples: a plant's height as it reaches full size; a car's speed as it approaches the speed limit; a learning curve.

Polynomials

W3. Zeros of $f(x) = (x-2)(x+5)$.

- A product is 0 when a factor is 0: $x-2=0$ or $x+5=0$.
- $\boxed{x=2,\ x=-5}$

W4. $f(x) = -2x^3 + x - 4$.

- Degree: highest power, **3**. Leading coefficient: **−2**.
- Odd degree with a negative leading coefficient: **up on the left, down on the right**.
- In limits: $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = -\infty$.

5. $f(x) = (x+2)(x-1)^2(x-3)$.

- (a) Set each factor to 0:

Zero	Factor	Multiplicity	Behavior
-2	$(x+2)^1$	1 (odd)	crosses
1	$(x-1)^2$	2 (even)	bounces
3	$(x-3)^1$	1 (odd)	crosses

- (b) Degree = sum of multiplicities $= 1+2+1 = 4$. Leading term: $x \cdot x^2 \cdot x = x^4$, so the leading coefficient is $+1$. Even degree, positive: **up on both ends**.
- (c) y-intercept: plug in $x=0$: $(2)(-1)^2(-3) = (2)(1)(-3) = \boxed{-6}$.

6. All zeros of $x^3 - x^2 + 4x - 4$.

- Group: $(x^3 - x^2) + (4x - 4) = x^2(x-1) + 4(x-1)$.
- Factor out $(x-1)$: $(x-1)(x^2+4)$.
- $x-1=0$ gives $x=1$.
- $x^2+4=0$ gives $x^2=-4$, so $x = \pm\sqrt{-4} = \pm 2i$.
- Zeros: $\boxed{1,\ 2i,\ -2i}$ (3 zeros for degree 3 ✓).

7. Degree 3, real coefficients, zeros 2 and $1+i$.

- Real coefficients mean complex zeros come in conjugate pairs, so $1-i$ is the third zero.
- Pair the complex factors first: $(x-(1+i))(x-(1-i))$. Group as $((x-1)-i)((x-1)+i)$, a difference of squares: $(x-1)^2 - i^2 = x^2 - 2x + 1 + 1 = x^2 - 2x + 2$.
- Multiply by $(x-2)$: $(x-2)(x^2-2x+2) = x^3 - 2x^2 + 2x - 2x^2 + 4x - 4$.
- Combine: $\boxed{x^3 - 4x^2 + 6x - 4}$

8. Even, odd or neither? Plug in $-x$.

- $f(-x) = (-x)^4 - 3(-x)^2 = x^4 - 3x^2 = f(x)$: **even**.
- $g(-x) = (-x)^3 + (-x) = -x^3 - x = -(x^3 + x) = -g(x)$: **odd**.
- $h(-x) = -x^3 + 1$. That's not $h(x) = x^3 + 1$, and not $-h(x) = -x^3 - 1$: **neither**.

9. End behavior of $p(x) = -3x^5 + 2x^2 - 1$.

- Leading term $-3x^5$: odd degree, negative coefficient, so up on the left and down on the right.
- $\lim_{x\to\infty} p(x) = -\infty$ and $\lim_{x\to-\infty} p(x) = \infty$

Rational functions

W5. $f(x) = \dfrac{1}{x-3}$.

- Vertical asymptote: the denominator is 0 at $x = 3$ and the top isn't, so $x = 3$.
- Horizontal asymptote: top degree 0, bottom degree 1. Bottom bigger, so $y = 0$.

10. $r(x) = \dfrac{2x^2 - 8}{x^2 - x - 6}$.

- Factor: top $2(x^2 - 4) = 2(x-2)(x+2)$; bottom $(x-3)(x+2)$.
- $(x+2)$ cancels, giving a **hole** at $x = -2$. Simplified: $\dfrac{2(x-2)}{x-3}$. Height of the hole: $\dfrac{2(-4)}{-5} = \dfrac{8}{5}$. Hole at $\left(-2, \frac{8}{5}\right)$.
- **Zero:** the remaining top factor, $x = 2$.
- **Vertical asymptote:** the remaining bottom factor, $x = 3$.
- **Horizontal asymptote:** degrees equal (2 and 2), so the ratio of leading coefficients: $y = \frac{2}{1} = 2$.
- **y-intercept:** $r(0) = \dfrac{-8}{-6} = \dfrac{4}{3}$.

11. $r(x) = \dfrac{x^2 + 1}{x - 1}$.

- Vertical asymptote: $x = 1$ (the top is $2 \neq 0$ there).
- The top's degree is one more than the bottom's, so there's a slant asymptote. Long division:
  - $x^2 \div x = x$. Multiply: $x(x-1) = x^2 - x$. Subtract: $(x^2 + 0x + 1) - (x^2 - x) = x + 1$.
  - $x \div x = 1$. Multiply: $1(x-1) = x - 1$. Subtract: $(x+1) - (x-1) = 2$.
  - So $r(x) = x + 1 + \dfrac{2}{x-1}$.
- As $x \to \pm\infty$, $\frac{2}{x-1} \to 0$, so the slant asymptote is $\boxed{y = x + 1}$.

12. End-behavior limits.

- $\dfrac{3x+1}{x^2+2}$: divide every term by x^2 to get $\dfrac{\frac{3}{x} + \frac{1}{x^2}}{1 + \frac{2}{x^2}} \to \dfrac{0}{1} = \boxed{0}$ (bottom degree bigger).
- $\dfrac{6x^2 - x}{2x^2 + 5}$: divide by x^2 to get $\dfrac{6 - \frac{1}{x}}{2 + \frac{5}{x^2}} \to \dfrac{6}{2} = \boxed{3}$ (equal degrees).

13. Solve $\dfrac{x-2}{x+1} \geq 0$.

- Critical points: top $= 0$ at $x = 2$; bottom $= 0$ at $x = -1$.
- Test each interval:

Interval	Test x	$\frac{x-2}{x+1}$	Sign
$x < -1$	-2	$\frac{-4}{-1} = 4$	$+$
$-1 < x < 2$	0	$\frac{-2}{1} = -2$	$-$
$x > 2$	3	$\frac{1}{4}$	$+$

- Include $x = 2$ (the expression equals 0 there, and $\geq$ allows it). Exclude $x = -1$ (division by 0).
- $\boxed{x < -1 \text{ or } x \geq 2}$

Equivalent forms

14. $(2x^3 + 3x^2 - x + 5) \div (x + 2)$.

- $2x^3 \div x = 2x^2$. Multiply: $2x^2(x + 2) = 2x^3 + 4x^2$. Subtract: $3x^2 - 4x^2 = -x^2$. Bring down: $-x^2 - x$.
- $-x^2 \div x = -x$. Multiply: $-x(x + 2) = -x^2 - 2x$. Subtract: $-x - (-2x) = x$. Bring down: $x + 5$.
- $x \div x = 1$. Multiply: $1(x + 2) = x + 2$. Subtract: $5 - 2 = 3$.
- $\boxed{2x^2 - x + 1 + \dfrac{3}{x+2}}$
- Check with synthetic division by -2: coefficients 2, 3, −1, 5 give 2, −1, 1 with remainder 3 ✓.

15. Expand $(x + 2)^4$.

- Pascal's row 4: 1, 4, 6, 4, 1. Powers of x count down from 4, and powers of 2 count up from 0.
- $1x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 1(16)$
- $\boxed{x^4 + 8x^3 + 24x^2 + 32x + 16}$

Transformations

W6. $g(x) = f(x - 3) + 2$.

- $x - 3$ inside: shift **right 3**. The $+2$ outside: shift **up 2**.

16. $g(x) = -2f(x + 1) - 4$ with $f(x) = x^2$.

- Inside $x + 1$: shift **left 1**.
- Times 2: **vertical stretch** by 2.
- Negative sign: **reflect over the x-axis**.
- -4: shift **down 4**.
- The vertex of x^2 is $(0, 0)$. Left 1 and down 4 move it to $\boxed{(-1, -4)}$. The reflection makes the parabola open down.

17. $(4, 6)$ is on f. Find a point on $g(x) = 3f(2x) - 1$.

- We know $f(4) = 6$, so we need the input to f to be 4: $2x = 4$, so $x = 2$.
- Then $g(2) = 3f(4) - 1 = 3(6) - 1 = 17$.
- Point: $\boxed{(2, 17)}$
- Pattern: the inside "×2" **halves** x-coordinates; the outside "×3, −1" acts directly on y.

Modeling

18. Quadratic $f(x) = ax^2 + bx + c$ through $(0, 5)$, $(1, 8)$, $(2, 15)$.

- $(0, 5)$: $c = 5$.
- $(1, 8)$: $a + b + 5 = 8$, so $a + b = 3$.
- $(2, 15)$: $4a + 2b + 5 = 15$, so $4a + 2b = 10$, which is $2a + b = 5$.
- Subtract the two equations: $(2a + b) - (a + b) = 5 - 3$, so $a = 2$. Then $b = 1$.
- $f(x) = 2x^2 + x + 5$, and $f(5) = 50 + 5 + 5 = \boxed{60}$.

Unit 2: Exponential and logarithmic functions

Sequences

W7. (a) $3, 7, 11, 15$: differences all $+4$, so **arithmetic,** $d = 4$. (b) $2, 6, 18, 54$: ratios all $\times 3$, so **geometric,** $r = 3$.

19. $a_1 = 5$, $d = 3$.

- $a_n = a_1 + d(n-1) = 5 + 3(n-1)$ (or simplified, $3n + 2$).
- $a_{20} = 5 + 3(19) = 5 + 57 = \boxed{62}$

20. $g_1 = 4$, $r = \frac{1}{2}$.

- $g_n = g_1 r^{n-1} = 4\left(\frac{1}{2}\right)^{n-1}$
- $g_6 = 4\left(\frac{1}{2}\right)^5 = \frac{4}{32} = \boxed{\frac{1}{8}}$

21. $a_3 = 11$, $a_8 = 26$.

- From term 3 to term 8 is 5 steps: $5d = 26 - 11 = 15$, so $d = 3$.
- Back up 2 steps from a_3: $a_1 = 11 - 2(3) = \boxed{5}$.

Linear vs. exponential

22.

- (a) $5, 10, 20, 40$: differences 5, 10, 20 (not constant). Ratios $\frac{10}{5} = \frac{20}{10} = \frac{40}{20} = 2$ (constant). **Exponential**: start 5, times 2: $\boxed{y = 5 \cdot 2^x}$.
- (b) $5, 8, 11, 14$: differences all 3. **Linear**: start 5, slope 3: $\boxed{y = 3x + 5}$.

23. $y = ab^x$ through two points.

- (a) $(0, 3)$: $a \cdot b^0 = a = 3$. $(2, 48)$: $3b^2 = 48$, so $b^2 = 16$ and $b = 4$ (it must be positive). $\boxed{y = 3 \cdot 4^x}$
- (b) $ab^1 = 6$ and $ab^3 = 24$. Divide the second by the first: $\dfrac{ab^3}{ab} = b^2 = \dfrac{24}{6} = 4$, so $b = 2$. Then $2a = 6$, so $a = 3$. $\boxed{y = 3 \cdot 2^x}$

Exponential functions and models

W8. (a) $b = 1.08 > 1$: **growth**, $1.08 - 1 = 0.08$, so **8%**. (b) $b = 0.75 < 1$: **decay**, $1 - 0.75 = 0.25$, so **25%**.

24. 2,000 people, growing 5% per year.

- Growth factor $1 + 0.05 = 1.05$: $P(t) = 2000(1.05)^t$
- $P(10) = 2000(1.05)^{10} \approx 2000(1.6289) \approx \boxed{3258 \text{ people}}$

25. \$24,000, losing 15% per year.

- Decay factor $1 - 0.15 = 0.85$: $V(t) = 24000(0.85)^t$
- $V(4) = 24000(0.85)^4 = 24000(0.52200625) \approx \boxed{\$12{,}528.15}$

26. Rewrites.

- (a) $3^{x+2} = 3^x \cdot 3^2 = \boxed{9 \cdot 3^x}$
- (b) $8 = 2^3$, so $8^x = (2^3)^x = \boxed{2^{3x}}$
- (c) $\frac{1}{2} = 2^{-1}$, so $\left(\frac{1}{2}\right)^x = (2^{-1})^x = \boxed{2^{-x}}$
- (d) 12 months make a year, so the monthly factor m satisfies $m^{12} = 1.2$, giving $m = 1.2^{1/12} \approx \boxed{1.0153}$, about 1.53% per month.

27. \$1,000 at 4% for 10 years.

- Continuous: $A = Pe^{rt} = 1000e^{0.04 \cdot 10} = 1000e^{0.4} \approx \boxed{\$1491.82}$
- Monthly: $A = P\left(1 + \frac{r}{n}\right)^{nt} = 1000\left(1 + \frac{0.04}{12}\right)^{120} \approx \boxed{\$1490.83}$
- Continuous is slightly more, as it should be: it's the limit of compounding more and more often.

Composition and inverses

W9. $f(x) = 2x + 1$, $g(x) = x^2$.

- $f(g(3))$: inside first, $g(3) = 9$. Then $f(9) = 19$.
- $g(f(3))$: $f(3) = 7$. Then $g(7) = 49$.

28.

- $f(g(x)) = 2(x^2) + 1 = 2x^2 + 1$
- $g(f(x)) = (2x + 1)^2 = 4x^2 + 4x + 1$
- **Not equal.** Order matters (W9 already showed it: 19 vs. 49).

29. Inverses.

- (a) $y = \frac{x-4}{3}$. Swap: $x = \frac{y-4}{3}$. Multiply by 3: $3x = y - 4$. So $\boxed{f^{-1}(x) = 3x + 4}$.
- (b) $y = 2^x + 1$. Swap: $x = 2^y + 1$. Subtract 1: $2^y = x - 1$. Rewrite as a log: $\boxed{f^{-1}(x) = \log_2(x - 1)}$.

30. $f(x) = x^3 - 2$, $g(x) = \sqrt[3]{x + 2}$.

- $f(g(x)) = \left(\sqrt[3]{x + 2}\right)^3 - 2 = (x + 2) - 2 = x$ ✓
- $g(f(x)) = \sqrt[3]{(x^3 - 2) + 2} = \sqrt[3]{x^3} = x$ ✓
- Both compositions give x, so they're inverses.

Logarithms

W10. Each log asks "what power?"

- $\log_2 8$: $2^? = 8$, so **3**.
- $\log 1000$: $10^? = 1000$, so **3**.
- $\ln e^5$: $e^? = e^5$, so **5**.
- $\log_3 1$: $3^? = 1$, so **0**.
- $\log_5 \frac{1}{25}$: $5^? = 5^{-2}$, so **−2**.

31.

- (a) $2^5 = 32$: the base stays the base and the exponent is the answer: $\boxed{\log_2 32 = 5}$.
- (b) $\log_4 x = 3$ means $4^3 = x$, so $\boxed{x = 64}$.

32. Expand $\log\left(\dfrac{x^3 y}{z^2}\right)$.

- Quotient rule: $\log(x^3 y) - \log(z^2)$.
- Product rule: $\log x^3 + \log y - \log z^2$.
- Power rule: $\boxed{3\log x + \log y - 2\log z}$.

33. Condense $2\ln x - \ln(x + 1) + \ln 3$.

- Power rule first: $2\ln x = \ln x^2$.
- Add the positive terms (product): $\ln x^2 + \ln 3 = \ln(3x^2)$.
- Subtract (quotient): $\boxed{\ln\left(\dfrac{3x^2}{x + 1}\right)}$.

34. $\log_3 20 = \dfrac{\ln 20}{\ln 3} = \dfrac{2.9957}{1.0986} \approx \boxed{2.727}$. Sanity check: $3^2 = 9$ and $3^3 = 27$, so the answer should be between 2 and 3 ✓.

35. $f(x) = \log_2(x-3) + 1$.

- Domain: the log's input must be positive: $x - 3 > 0$, so $\boxed{x > 3}$.
- Vertical asymptote where the input is 0: $\boxed{x = 3}$.
- x-intercept: set $f(x) = 0$. Then $\log_2(x-3) = -1$, so $x - 3 = 2^{-1} = \frac{1}{2}$, and $\boxed{x = 3.5}$.

Equations and inequalities

W11. $2^x = 32 = 2^5$, so $x = 5$.

36. $3^{2x-1} = 27$.

- Same base: $27 = 3^3$, so $3^{2x-1} = 3^3$.
- Set the exponents equal: $2x - 1 = 3$, so $2x = 4$ and $\boxed{x = 2}$.

37. $5^x = 40$.

- 40 isn't a power of 5, so take ln of both sides: $\ln(5^x) = \ln 40$.
- Power rule: $x \ln 5 = \ln 40$, so $x = \dfrac{\ln 40}{\ln 5} \approx \boxed{2.292}$.
- Check: $5^2 = 25$ and $5^3 = 125$, so the answer is between 2 and 3 ✓.

38. $4e^{2x} = 20$.

- Isolate the exponential first: $e^{2x} = 5$.
- Take ln: $2x = \ln 5$, so $x = \dfrac{\ln 5}{2} \approx \boxed{0.805}$.

39. $\log_2 x + \log_2(x-2) = 3$.

- Condense with the product rule: $\log_2[x(x-2)] = 3$.
- Rewrite as an exponential: $x(x-2) = 2^3 = 8$.
- $x^2 - 2x - 8 = 0$, which factors as $(x-4)(x+2) = 0$: $x = 4$ or $x = -2$.
- Check: $x = -2$ gives $\log_2(-2)$, which is undefined, so **reject it**. $x = 4$: $\log_2 4 + \log_2 2 = 2 + 1 = 3$ ✓.
- $\boxed{x = 4}$

40. $\ln(x+1) = 2$ means $e^2 = x + 1$, so $x = e^2 - 1 \approx \boxed{6.389}$.

41. $2^x > 100$.

- Take $\log_2$ of both sides (base $2 > 1$, so the inequality keeps its direction): $x > \log_2 100 = \dfrac{\ln 100}{\ln 2} \approx \boxed{6.644}$.
- Check: $2^6 = 64 < 100 < 128 = 2^7$ ✓.

42. When does $2000(1.05)^t = 5000$?

- Divide by 2000: $1.05^t = 2.5$.
- Take ln: $t \ln 1.05 = \ln 2.5$, so $t = \dfrac{\ln 2.5}{\ln 1.05} \approx \boxed{18.8 \text{ years}}$.

Modeling and semi-log plots

43. 80 mg, half-life 6 hours.

- It halves every 6 hours, so the exponent counts half-lives: $A(t) = 80\left(\frac{1}{2}\right)^{t/6}$.
- $A(15) = 80\left(\frac{1}{2}\right)^{2.5} \approx 80(0.1768) \approx \boxed{14.1 \text{ mg}}$
- Sanity check: after 12 hours (2 half-lives) it's 20 mg, and after 18 hours it's 10 mg. 14.1 is between ✓.

44. $\log y = 0.3x + 1$.

- Undo the log (base 10): $y = 10^{0.3x+1}$.
- Split the exponent: $y = 10^1 \cdot 10^{0.3x} = 10\left(10^{0.3}\right)^x$.
- $10^{0.3} \approx 1.995$, so $\boxed{y \approx 10(1.995)^x}$, which is about $10 \cdot 2^x$.

45. Start with $y = ab^x$ and take log of both sides:

- $\log y = \log(ab^x) = \log a + \log b^x = \log a + x \log b$.
- That's $\log y = (\log b)x + \log a$, the form $Y = mx + c$ with $Y = \log y$.
- So plotting $\log y$ against x gives a straight line, with slope $\log b$ and intercept $\log a$.